A Step-by-Step Derivation of the Repaired Tether Rate: Identities that Close, Inequalities that Remain Open

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Abstract

We unfold the Lean proposition `FrohmanianTetherRateSmooth` and derive, step by step, the evolution of $M(t) = \|\omega(t)\|_\infty$ along a local solution in Fefferman's smooth class on $\mathbb{R}^3 \times [0, T)$. Every calculus identity that closes is kernel-checked in Lean 4 (`v4.30.0-rc1`, Mathlib rev `0c154d67`), using only the axioms `propext`, `Classical.choice` and `Quot.sound`, with no `sorry`. Every inequality that does not close is written as a named `Prop`, tested against Navier–Stokes parabolic scaling, and compared with known counterexamples. The statement of `FrohmanianTetherRateSmooth` is unchanged, and it is not proved for general Fefferman data. This document does not resolve any Clay Millennium Problem.

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1 The statement, unfolded

In the repository `frohmanian-tether-sdiff`, on the branch used for this derivation, the repaired rate is `RepairedTether.FrohmanianTetherRateSmooth` in `Scratch/RepairedTetherRate.lean:72--79`. It reads, verbatim:

The definitions it unfolds to, quoted verbatim from `Scratch/RepairedTetherRate.lean:54--79`, are:

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```
@[expose] public def IsFeffermanSolutionOn (ν T : ℝ) (u₀ : VelocityField)
    (u : TimeDependentVelocity) (p : TimeDependentPressure) : Prop :=
  ContDiffOn ℝ ∞ (fun q : T3 × ℝ => u q.2 q.1) (Set.univ ×ˢ Set.Ico (0 : ℝ) T) ∧
  ContDiffOn ℝ ∞ (fun q : T3 × ℝ => p q.2 q.1) (Set.univ ×ˢ Set.Ico (0 : ℝ) T) ∧
  (∀ t ∈ Set.Ico (0 : ℝ) T, ∀ x : T3,
    derivWithin (fun s => u s x) (Set.Ici 0) t + convective (u t) (u t) x +
        pressureGradient (p t) x = ν • laplacian (u t) x + 0 ∧
      div (u t) x = 0) ∧
  (∀ x : T3, u 0 x = u₀ x) ∧
  (∀ t ∈ Set.Ico (0 : ℝ) T, ∫⁻ x, ‖u t x‖ₑ ^ 2 ≤ ∫⁻ x, ‖u₀ x‖ₑ ^ 2)

@[expose] public def Mω (u : TimeDependentVelocity) (t : ℝ) : ℝ :=
  vorticity_sup_norm (vorticity (u t))

@[expose] public def FrohmanianTetherRateSmooth (u₀ : VelocityField) (ν : ℝ) : Prop :=
  ∃ C kap : ℝ, 0 < C ∧ 0 < kap ∧
    ∀ (T : ℝ) (u : TimeDependentVelocity) (p : TimeDependentPressure),
      IsFeffermanSolutionOn ν T u₀ u p →
        ContinuousOn (Mω u) (Set.Ico (0 : ℝ) T) ∧
        (∀ s ∈ Set.Ico (0 : ℝ) T, ∃ d : ℝ,
          HasDerivWithinAt (Mω u) d (Set.Ici s) s ∧
          d ≤ C * Mω u s ^ 2 - kap * Mω u s ^ 3) ∧
        (∀ t ∈ Set.Ico (0 : ℝ) T, BddAbove (Set.range fun x => ‖vorticity (u t) x‖))
```

The auxiliary pieces, all definitional, are as follows.

$M(t)$. By definition $M(t) = \|\omega(t)\|_\infty$, written in Lean as `Mω u t := vorticity_sup_norm (vorticity (u t))` with `vorticity_sup_norm omega := iSup fun x => ||omega x||` (`NS_Equations.lean:2787`). This is a real supremum. On an unbounded range Mathlib sets it to 0, which is why the rate carries an explicit `BddAbove` conjunct.

The solution class. `IsFeffermanSolutionOn nu T u0 u p` (`RepairedTetherRate.lean:54--63`) requires: joint smoothness of (u, p) on $\mathbb{R}^3 \times [0, T)$; the incompressible Navier–Stokes equations with $f \equiv 0$ and a one-sided time derivative at $t = 0$; the initial condition $u(\cdot, 0) = u_0$; and the energy inequality $\int_{\mathbb{R}^3} |u(t)|^2 \le \int_{\mathbb{R}^3} |u_0|^2$.

What must be shown. For every Fefferman datum u_0 and every $\nu > 0$, there exist constants $C, \kappa > 0$, depending only on u_0 and ν, such that along *every* local Fefferman-class solution issuing from u_0, the function M is finite and continuous on $[0, T)$, is right-differentiable, and obeys

$$M'(t) \;\le\; C\, M(t)^2 - \kappa\, M(t)^3. \tag{1}$$

The zero datum is already proved: `frohmanianTetherRateSmooth_zero`. For general data the proposition is unsettled. The original (unrepaired) rate `FrohmanianTetherRate` is kernel-checked disproved, as written, by `not_forall_frohmanianTetherRate`.

The Lean development of the identities below lives in `Scratch/TetherRateDerivation.lean` on the branch `feat/tether-rate-derivation`.

2 The evolution identity

Start from the vorticity formulation of Navier–Stokes, obtained by taking the curl of the momentum equation. For a two-sided solution this is the existing kernel-checked identity `vorticity_transport_at` (`NS_Equations.lean:2162`):

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega. \tag{2}$$

On a Fefferman solution the time derivative in the momentum equation is one-sided at $t = 0$. For times $t \in (0, T)$ the one-sided derivative coincides with the two-sided derivative, because $[0, \infty)$ is a neighbourhood of t. This is `fefferman_two_sided_eq_interior` (`TetherRateDerivation.lean:431`), together with `derivWithin_Ici_eq_deriv_of_pos` and the interior smoothness lemmas `fefferman_u_contDiffAt_interior`, `fefferman_p_contDiffAt_interior`. Status: **kernel-checked proved**. At $t = 0$ the transport identity must be read with a right derivative; that one-sided commutation of curl and ∂_t is not formalized in this file.

Take the inner product of (2) with ω and use the identity

$$\partial_t\Big(\tfrac{1}{2}|\omega|^2\Big) = \langle \omega, \partial_t \omega \rangle,$$

which is `hasDerivAt_halfNormSq_time`. One obtains the pointwise energy law

$$\partial_t\Big(\tfrac{1}{2}|\omega|^2\Big) + \langle \omega, (u \cdot \nabla)\omega \rangle = \langle \omega, (\omega \cdot \nabla)u \rangle + \nu \langle \omega, \Delta \omega \rangle. \tag{3}$$

Bochner identity. Write $|\nabla \omega|^2_{\mathrm{HS}} = \sum_{i=1}^{3} |\partial_i \omega|^2$ (`hsSq`) and write Δ for the scalar Laplacian (`scalarLaplacian`). For a C^2 field,

$$\langle \omega, \Delta \omega \rangle = \Delta\big(\tfrac{1}{2}|\omega|^2\big) - |\nabla \omega|^2_{\mathrm{HS}}. \tag{4}$$

This is `bochner_identity` (`TetherRateDerivation.lean:78`). Status: **kernel-checked proved**.

At a spatial maximum. Suppose x is a local spatial maximum of $y \mapsto |\omega(y)|$. Then it is a local maximum of $\frac{1}{2}|\omega|^2$ (`isLocalMax_halfNormSq_of_isLocalMax_norm`), the Fréchet derivative of $\frac{1}{2}|\omega|^2$ vanishes (`fderiv_halfNormSq_eq_zero_of_isLocalMax`), and therefore the transport pairing vanishes:

$$\langle \omega, (u \cdot \nabla)\omega \rangle(x) = 0. \tag{5}$$

This is `transport_vanishes_at_local_max` (`TetherRateDerivation.lean:147`), recovering the existing pointwise fact `transport_inner_vanishes_of_grad_zero`. Moreover, the Laplacian of a C^2 scalar is nonpositive at a local maximum (`scalarLaplacian_nonpos_of_isLocalMax`, `TetherRateDerivation.lean:202`, from the one-dimensional second-derivative test). Combined with (4),

$$\langle \omega, \Delta \omega \rangle(x) \;\le\; -|\nabla \omega(x)|^2_{\mathrm{HS}} \;\le\; 0. \tag{6}$$

This is `viscPair_le_neg_hsSq_at_local_max` (`TetherRateDerivation.lean:256`). Status of (5)–(6): **kernel-checked proved**.

Envelope for M. If $M(s) = |\omega(s, x(s))|$ along a differentiable maximizing trajectory, and if the spatial gradient of $\frac{1}{2}|\omega|^2$ vanishes at $x(t)$, the chain rule and (5) give, whenever $M(t) > 0$,

$$M'(t) = \frac{\langle \omega, \partial_t \omega \rangle}{M(t)}(t, x(t)). \tag{7}$$

This is `envelope_along_differentiable_maximizer` (`TetherRateDerivation.lean:317`), using `hasDerivAt_norm_of_ne_zero`. Status: **kernel-checked proved**, conditionally on the existence of such a path and on the splitting of the path derivative into $\partial_t \omega + (x' \cdot \nabla)\omega$. Existence of a (differentiable) maximizer on $\mathbb{R}^3$ is not implied by `IsFeffermanSolutionOn` as written, and is recorded as the named propositions `VorticityMaximumAttained` and `DifferentiableMaximizer`. Both are **unsettled**.

A difference-quotient form of Danskin's inequality, using a maximizer at time $t + h$ rather than at time t, is `danskin_difference_quotient` (`TetherRateDerivation.lean:351`). Status:

kernel-checked proved. The comparison that uses only a maximizer at time t goes the wrong way and does not bound M' from above.

Combining (3), (5), (6) and (7) at a spatial maximum with $M > 0$ yields

$$M' \;\le\; \frac{\langle \omega, (\omega\cdot\nabla)u\rangle + \nu\langle\omega,\Delta\omega\rangle}{M} \;\le\; \frac{\langle \omega, (\omega\cdot\nabla)u\rangle}{M} - \nu\frac{|\nabla\omega|^2_{\mathrm{HS}}}{M}. \tag{8}$$

3 Bounding the two terms

3.1 The stretching term, aiming at CM^2

By Cauchy–Schwarz and the definition of the operator-norm strain proxy `strain_sup_norm`,

$$\langle \omega, (\omega\cdot\nabla)u\rangle \;\le\; \|\nabla u\|_\infty\,|\omega|^2 \;\le\; \|\nabla u\|_\infty\, M^2,$$

at any point, once the strain is bounded. This is the existing identity `inner_convective_le_strain`, wrapped as `stretching_le_strain_mul_norm_sq` (`TetherRateDerivation.lean:363`). Status: **kernel-checked proved** (linear algebra, not a Calderón–Zygmund estimate).

To pass from this to a source CM^3, and hence after division by M to the quadratic term CM^2 in (1), one needs a bound of the form

$$\|\nabla u(t)\|_\infty \;\le\; C\,M(t)$$

uniformly in $t \in [0,T)$, with C depending only on u_0 and ν. This is the named proposition `StrainControlledByVorticity` (`TetherRateDerivation.lean:465`). Status: **unsettled** along Fefferman-class solutions.

A *universal* pointwise bound $\|\nabla u(x)\| \le C\|\omega\|_\infty$, holding for every (not necessarily decaying, not necessarily Biot–Savart) smooth field and a constant C independent of the field, is kernel-checked disproved by `StrainBoundRefutation.not_forall_strain_le_sup_vorticity`. The counterexample is an irrotational linear pure-strain field: it has $\omega = 0$ and $\nabla u \neq 0$. That field does not decay and is not recovered from vorticity by the Biot–Savart law, so the refutation does not by itself disprove `StrainControlledByVorticity` on Fefferman snapshots. On decaying divergence-free fields the map $\omega \mapsto \nabla u$ is a Calderón–Zygmund operator, which is unbounded on L^∞; the standard substitute is a logarithmic estimate involving a higher norm. Whether a constant $C = C(u_0,\nu)$ nevertheless controls $\|\nabla u\|_\infty/M$ uniformly in time along every local Fefferman solution is unsettled.

3.2 The dissipative term, aiming at $-\kappa M^3$

From (6), the viscous contribution to M' is at most $-\nu|\nabla\omega|^2_{\mathrm{HS}}/M$. At a spatial maximum of $|\omega|$, the polar decomposition $\nabla\omega = (\nabla|\omega|)\otimes\xi + |\omega|\nabla\xi$ and $\nabla|\omega| = 0$ give $|\nabla\omega|^2_{\mathrm{HS}} = M^2|\nabla\xi|^2_{\mathrm{HS}}$ (the algebraic Pythagoras identities `polar_pythagoras_directional_deriv` and `concrete_bochner_polar_viscous_damping_at_max` in `SymplecticConstraint.lean` already record this reduction). The cubic sink $-\kappa M^3$ in (1) is then equivalent, after dividing by M, to a lower bound

$$|\nabla\omega|^2_{\mathrm{HS}} \;\ge\; \kappa\, M^4 \qquad \text{i.e.} \qquad |\nabla\xi|^2_{\mathrm{HS}} \;\ge\; \kappa\, M^2 \tag{9}$$

at the attained maximum, with $\kappa > 0$ independent of t. This is the named proposition `DirectionCoercivityAtMax` (`TetherRateDerivation.lean:476`). Status along Navier–Stokes trajectories: **unsettled**.

The same lower bound, imposed on arbitrary smooth snapshots rather than on a Navier–Stokes trajectory, is not a general fact about maxima of $|\omega|$. A rigid rotation $u(x) = (-x_2, x_1, 0)$ has constant vorticity $\omega = (0,0,2)$, so every point is a maximum, $M = 2$, and $|\nabla\omega|_{\mathrm{HS}} = 0$.

The dissipation term vanishes identically, and (9) fails for every $\kappa > 0$. This field does not decay and is not a Fefferman datum; the example shows only that (9) cannot be a universal static inequality, so if it holds along a solution it must use the dynamics. A wide, nearly rigid Gaussian vortex $u(x) = e^{-\varepsilon|x|^2}(-x_2, x_1, 0)$ produces the same vanishing of $|\nabla\omega|$ at the origin and a dissipation of size $O(\varepsilon)$, with $M = 2$ fixed as $\varepsilon \to 0$. That calculation is recorded here as an analytic test, not as a kernel-checked Lean theorem in this commit.

3.3 Algebraic assembly

Once stretching is at most CM^3 and $\nu\langle\omega, \Delta\omega\rangle$ is at most $-\kappa M^4$, division by $M > 0$ yields (1). This is `cubic_rate_of_stretching_and_visc` and `cubic_rate_of_strain_and_hsSq` (`TetherRateDerivation.lean:373` and `:390`), and the hook `cubic_rate_from_named_hypotheses`. Status: **kernel-checked proved** as an implication. It does not prove the two hypotheses, and it does not prove `FrohmanianTetherRateSmooth`.

4 Scaling analysis

The Navier–Stokes scaling is $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$, formalized as `rescaleFlow`. The vorticity scales as $\omega_\lambda(x,t) = \lambda^2\omega(\lambda x, \lambda^2 t)$ (`vorticity_rescaleFlow`, extending the existing spatial identity `ScalingIdentities.vorticity_rescale`). Consequently, for $\lambda \neq 0$,

$$M(u_\lambda, t) = |\lambda|^2\, M(u, \lambda^2 t),$$

which is `Mω_rescaleFlow` (`TetherRateDerivation.lean:515`). If M is differentiable at $\lambda^2 t$ and $\lambda > 0$, the rescaled ceiling has derivative $\lambda^4 M'(\lambda^2 t)$ (`hasDerivAt_Mω_rescale_of_hasDerivAt`).

The form of the proposed rate is not homogeneous of a single degree. Writing $s = \lambda^2 t$ and $M_\lambda(\tau) = \lambda^2 M(\lambda^2\tau)$, the identity

$$\lambda^4 M'(s) \;\le\; C\, M_\lambda(t)^2 - \kappa\, M_\lambda(t)^3 \quad\Longleftrightarrow\quad M'(s) \;\le\; C\, M(s)^2 - \kappa\lambda^2 M(s)^3$$

is `rate_form_rescale_iff` (`TetherRateDerivation.lean:568`). Status: **kernel-checked proved**.

The quadratic term CM^2 scales as the time derivative of M (degree λ^4). The cubic sink $-\kappa M^3$ scales as λ^6. It is therefore one degree supercritical for Navier–Stokes scaling: a pair (C, κ) that is independent of the concentration parameter λ is not preserved under rescaling of a single profile. This does not disprove `FrohmanianTetherRateSmooth`, whose constants are allowed to depend on u_0. It does constrain that dependence: for data concentrated at scale λ^{-1}, any admissible κ must shrink at least as λ^{-2}.

The kinetic energy of the rescaled field is λ^{-1} times the original energy, which is the usual statement that the three-dimensional energy is supercritical relative to the scaling of $\|\omega\|_\infty$.

5 What is proved, what is disproved, and what is unsettled

Claim	Status	Lean name, file:line
Bochner identity (4)	proved	`bochner_identity`, :78
Transport cancels at a local max	proved	`transport_vanishes_at_local_max`, :147
$\Delta(\frac{1}{2}\lvert\omega\rvert^2) \le 0$ at a local max	proved	`scalarLaplacian_nonpos_of_isLocalMax`, :202
$\langle\omega, \Delta\omega\rangle \le -\lvert\nabla\omega\rvert^2_{\mathrm{HS}}$ at a local max	proved	`viscPair_le_neg_hsSq_at_local_max`, :256
Envelope (7) along a C^1 maximizer	proved, conditional on a path	`envelope_along_differentiable_maximizer`, :317
Danskin difference quotient	proved	`danskin_difference_quotient`, :351
Stretching $\le \lVert\nabla u\rVert_\infty M^2$	proved	`stretching_le_strain_mul_norm_sq`, :363
Two-sided NS equation on $(0,T)$	proved	`fefferman_two_sided_eq_interior`, :431
Algebraic assembly into (1)	proved as an implication	`cubic_rate_of_strain_and_hsSq`, :390
M and M' under parabolic scaling	proved	`Mω_rescaleFlow`, :515; `rate_form_rescale_iff`, :568
Universal $\lVert\nabla u\rVert \le C\lVert\omega\rVert_\infty$ for arbitrary fields	disproved	`not_forall_strain_le_sup_vorticity` in `StrainBoundRefutation`
Existence of a spatial maximizer on $\mathbb{R}^3$	unsettled	`VorticityMaximumAttained`, :448
Differentiable maximizing path on $(0,T)$	unsettled	`DifferentiableMaximizer`, :454
Danskin rate without a C^1 path	unsettled	`DanskinRateAt`, :485
$\lVert\nabla u\rVert_\infty \le C(u_0,\nu)\,M$ along solutions	unsettled	`StrainControlledByVorticity`, :465
$\lvert\nabla\omega\rvert^2_{\mathrm{HS}} \ge \kappa(u_0,\nu)\,M^4$ along solutions	unsettled	`DirectionCoercivityAtMax`, :476
`FrohmanianTetherRateSmooth` for general data	unsettled	statement unchanged, `RepairedTetherRate.lean:72`
`FrohmanianTetherRateSmooth` at $u_0 = 0$	proved (pre-existing)	`frohmanianTetherRateSmooth_zero`
`FrohmanianTetherRate` as originally written	disproved (pre-existing)	`not_forall_frohmanianTetherRate`

6 Toolchain, axioms, and rebuild

The identities were checked with Lean `v4.30.0-rc1` and Mathlib revision `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7` (`v4.30.0-rc1`), on the branch `feat/tether-rate-derivation` of `github.com/BenFrohman/frohmanian-tether-sdiff`. Every printed theorem in `Scratch/TetherRateDerivation.lean` depends only on `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` in the file. Rebuild with

```
lake build Scratch.TetherRateDerivation
```

from a checkout of that branch, after the Mathlib package at the pinned revision is present. The statement of `FrohmanianTetherRateSmooth` was not changed, and this work does not prove it for general Fefferman data.